CANONICAL TRANSFORMATIONS DEPENDING ON A SMALL PARAMETER

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Abstract. The concept of a Lie series is enlarged to encompass the cases where the generating function itself depends explicitly on the small parameter. Lie transforms define naturally a class of canonical mappings in the form of power series in the small parameter. The formalism generates nonconservative as well as conservative transformations. Perturbation theories based on it offer three substantial advantages: they yield the transformation of state variables in an explicit form; in a function of the original variables, substitution of the new variables consists simply of an iterative procedure involving only explicit chains of Poisson brackets; the inverse transformation can be built the same way.

1. Introduction

Canonical transformations developed in formal series

$$x = y + \sum_{n\geqslant 1} \frac{1}{n!}\varepsilon^n y^{(n)}(y, Y; t),$$

$$X = Y + \sum_{n\geqslant 1} \frac{1}{n!}\varepsilon^n Y^{(n)}(y, Y; t) \tag{1}$$

in the powers of a small parameter ε play an important role in the quantitative as well as qualitative study of dynamical systems described by Hamiltonians of the type

$$\mathscr{H} \equiv \mathscr{H}(x, X; t; \varepsilon) = \sum_{n\geqslant 0} \frac{1}{n!}\varepsilon^n \mathscr{H}_n(x, X; t). \tag{2}$$

When the transformations are conservative, the so-called von Zeipel's method proposes to build them from generating functions in mixed variables, like

$$W \equiv W(x, Y; \varepsilon) = (x|Y) + \sum_{n\geqslant 1} \frac{1}{n!}\varepsilon^n W_n(x, Y),$$

by the implicit equations

$$X \equiv X(x, Y; \varepsilon) = W_x, \tag{3}$$

$$y \equiv y(x, Y; \varepsilon) = W_Y. \tag{4}$$

(N.B. $(x|Y)$ denotes the dot product of the configuration vector x and the moment vector Y; W_x indicates the gradient of W in the configuration space x, a vector made of the partial derivatives of W with respect to the components of x; W_Y has a similar interpretation.)

Von Zeipel's method presents serious inconveniences:

(a) An explicit formulation of the transformation and of its inverse requires inverting the Equations (4) to obtain the configuration vector x as a function of the state variables (y, Y), whereafter the result of the inversion is substituted into the Equations (3) to express explicitly the moment vector X in terms of the state variables (y, Y).

From a practical viewpoint, inversions and substitutions are cumbersome operations. Oftentimes the authors of a perturbation theory are satisfied with suggesting a Lagrange series extended to several variables in order to invert Equation (4), but they do not enter into the details of the operations. As a result, their theory is left unfinished, precisely on a point to which practitioners ought to pay special care. For instance, ephemerides of an artificial satellite based on a theory elaborated by means of a von Zeipel's technique may suffer in accuracy from the fact that Delaunay's elements averaged over an anomalistic period are not explicitly developed in terms of the initial conditions (see Lyddane and Cohen, 1962).

(b) The implicit relations (3) and (4) do not lead to a general algorithm to convert into the state variables (y, Y) a function

$$f(x, X : \varepsilon) = \sum_{n \geq 0} \frac{1}{n!} \varepsilon^n f_n(x, X)$$

of the original state variables (x, X). But these conversions have a considerable importance in practice. In the sequence of computations needed to establish and differentially correct the orbit of an artificial satellite, substantial savings would be made if the perturbation theories would express explicitly state functions like longitude, latitude, range and range rate in terms of the averaged elements.

(c) In numerous problems, a transformation to be appropriate must satisfy, besides the criterion of canonicity, other essential conditions, like being analytical in the moments Y or defining the state variables (x, X) as averages of their transformed ones (y, Y). Some of these properties easily transfer from the mapping to the generating function. But this is not the general case. How, for instance, do we select those functions W such that, after inversion and substitution, the coefficients $y^{(n)}$ and $Y^{(n)}$ be analytical in the moment vector Y? The problem is a real one. According to Barton (1967), it is a major obstacle to implementing by computer a Delaunay operation.

We felt the shortcomings of von Zeipel's method when we attempted to apply it to normalize a Hamiltonian system in the neighborhood of an equilibrium (Deprit, 1966). As a result of this failure, we found a way of constructing the transformation (1) in a straightforward recursion based on an extension to quasi-periodic solutions of Poincaré's continuation of periodic orbits (Deprit and Henrard, 1967). However, we still failed to give a good answer to the problem of converting in the normalizing variables a function of the primitive state variables (x, X).

In the present communication, we offer a Theory of Transformations that, in our opinion, accomplishes everything that von Zeipel's method does in a general perturbation theory, and yet remedies its shortcomings and inconveniences.

For the sake of clarity in the exposition, we would like to present the basic idea as a generalization of the concept of an infinitesimal canonical transformation. We proceed in two stages. First we review how a Lie series defines a canonical mapping as a formal power series of a small parameter ε, provided the generating function itself does not depend on ε. Then we overcome this restriction by introducing the concept of a Lie transform, and we show how they naturally define the canonical transformations contemplated by von Zeipel's method. We complete this paper by outlining the basic steps of a general theory of perturbations based on Lie transforms.

A little consideration of the past literature makes it apparent that the class of canonical transformations introduced in this communication has been envisioned by several authors.

(a) It naturally encompasses the *transition* mappings from initial states at t_0 to the states at a nearby instant t as they are determined by the general solution of a conservative Hamiltonian system. Gröbner (1960) recommends presenting the transition mappings as Lie series generated by the Hamiltonian itself; recently Stumpff (1968) commented on possible applications of Gröbner's formalism. In that context, Section 3 of this paper offers a generalization of Lie series to nonconservative Hamiltonian systems (see also Leimanis, 1965).

(b) Von Zeipel's technique and the present one are identical to the first order. At this approximation, the conversion rule applied to a function that does not depend explicitly on the small parameter has been proposed by Iszak (1963). As a matter of fact, Iszak's formula was inspired by a remark of Dziobek (1888) that purports to justify a similar operation in the first order of a planetary theory.

(c) The construction that we propose here ought to be discussed in relation with an algorithm already proposed by Hori (1966, 1967). Although the preamble to his construction seems amenable to a serious objection, a thorough comparison of the two formalisms could prove very informative.

(d) The present formalism justifies the way Schechter (1967) corrected a normalization around an equilibrium carried by Breakwell and Pringle (1966) in the case of a resonance. What Schechter does there amounts to deriving from a von Zeipel's generating function in mixed variables the Lie transforms that define the transformation in explicit form.

(e) Most recently, Lacina (1969) has proposed a recursive algorithm to construct explicitly a canonical transformation that renders all variables, except a moment, ignorable in a conservative Hamiltonian function. Lacina's mapping is not derived from a generating function.

2. Lie Series as Canonical Mappings

Given a function $W(y, Y)$, the application $f \to (f; W)$, where

$$(f; W) = \sum_{1 \leqslant j \leqslant m} \frac{\partial f}{\partial y_j}\frac{\partial W}{\partial Y_j} - \frac{\partial f}{\partial Y_j}\frac{\partial W}{\partial y_j}$$

is the Poisson bracket of f and W over the phase space (y, Y), is called the *Lie derivative generated by* W, and it is denoted by L_W.

For any real numbers α and β, and for any functions $f(y, Y)$ and $g(y, Y)$,

$$L_W(\alpha \cdot f + \beta \cdot g) = \alpha \cdot L_W f + \beta \cdot L_W g\,, \tag{1$_1$}$$

$$L_W(f \cdot g) = f \cdot L_W g + g \cdot L_W f\,, \tag{2$_1$}$$

$$L_W(f; g) = (f; L_W g) + (L_W f; g)\,; \tag{3$_1$}$$

also, for any function $V(y, Y)$,

$$L_V L_W = L_{(W;V)} + L_W L_V\,. \tag{4$_1$}$$

These properties are obvious transpositions of corresponding properties of Poisson brackets.

Repeated applications of the operator L_W give

$$L_W^2 f = L_W(L_W f), \quad L_W^3 f = L_W(L_W^2 f),$$

and so on. It makes for convenient notations to put

$$L_W^0 f = f \quad \text{and} \quad L_W^1 f = L_W f\,.$$

Notice then how, by recurrence over n, properties (1), (2) and (3) can be extended to become

$$L_W^n(\alpha f + \beta g) = \alpha L_W^n f + \beta L_W^n g\,, \tag{1$_n$}$$

$$L_W^n(f \cdot g) = \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_W^m f \cdot L_W^{n-m} g\,, \tag{2$_n$}$$

$$L_W^n(f; g) = \sum_{0 \leqslant m \leqslant n} \binom{n}{m} (L_W^m f; L_W^{n-m} g)\,. \tag{3$_n$}$$

Let the function $W(y, Y)$ satisfy conditions sufficient to ensure that the exponential operator

$$\exp(\varepsilon L_W) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n L_W^n \tag{5}$$

exists at any point (y, Y) of the phase space as an operator function analytic in ε at $\varepsilon = 0$. The following properties are immediate consequences of the corresponding formulas (1$_n$), (2$_n$) and (3$_n$):

$$\exp(\varepsilon L_W)(\alpha f + \beta g) = \alpha \cdot \exp(\varepsilon L_W) f + \beta \cdot \exp(\varepsilon L_W) g\,, \tag{1$_e$}$$

$$\exp(\varepsilon L_W)(f g) = [\exp(\varepsilon L_W) f] \times [\exp(\varepsilon L_W) g]\,, \tag{2$_e$}$$

$$\exp(\varepsilon L_W)(f; g) = (\exp(\varepsilon L_W) f; \exp(\varepsilon L_W) g)\,. \tag{3$_e$}$$

Formula (3$_e$) obviously implies that the transformation $(y, Y, \varepsilon) \to (x, X)$ defined by the formulas

$$x = \exp(\varepsilon L_W)(y), \quad X = \exp(\varepsilon L_W)(Y) \tag{6}$$

is completely canonical. For it establishes that

$$(x_j;\, x_k) = \exp(\varepsilon L_W)\,(y_j;\, y_k) = \exp(\varepsilon L_W)\cdot 0 = 0\,,$$
$$(x_j;\, X_k) = \exp(\varepsilon L_W)\,(y_j;\, Y_k) = \exp(\varepsilon L_W)\cdot \delta_{jk} = \delta_{jk}\,,$$
$$(X_j;\, X_k) = \exp(\varepsilon L_W)\,(Y_j;\, Y_k) = \exp(\varepsilon L_W)\cdot 0 = 0\,.$$

Take an indefinitely differentiable function $f(x, X)$ that does not depend explicitly on ε, and substitute in it for x and X their expressions given by (6)

$$\bar{f}(y,\, Y;\, \varepsilon) = f\big(x(y,\, Y;\, \varepsilon),\, X(y,\, Y;\, \varepsilon)\big)\,.$$

One computes that

$$(\bar{f};\, W) = \sum_{1\leqslant k\leqslant n}\left[(x_k;\, W)\frac{\partial f}{\partial x_k} + (X_k;\, W)\frac{\partial f}{\partial X_k}\right]$$

which, of course, can be written as

$$L_W\bar{f} = (f_x \,|\, L_W x) + (f_X \,|\, L_W X)\,.$$

But

$$\frac{\mathrm{d}x}{\mathrm{d}\varepsilon} = \sum_{n\geqslant 0}\frac{1}{n!}\,\varepsilon^n L_W^{n+1} y = L_W x\,,$$

and, similarly

$$\frac{\mathrm{d}X}{\mathrm{d}\varepsilon} = L_W X\,.$$

Consequently

$$(\bar{f};\, W) = \sum_{1\leqslant k\leqslant n}\left(\frac{\partial f}{\partial x_k}\frac{\mathrm{d}x_k}{\mathrm{d}\varepsilon} + \frac{\partial f}{\partial X_k}\frac{\mathrm{d}X_k}{\mathrm{d}\varepsilon}\right) = \frac{\mathrm{d}\bar{f}}{\mathrm{d}\varepsilon}\,.$$

Hence, by recurrence over n,

$$\frac{\mathrm{d}^n\bar{f}}{\mathrm{d}\varepsilon^n} = L_W^n\bar{f}\,,$$

which implies that

$$\left(\frac{\mathrm{d}^n\bar{f}}{\mathrm{d}\varepsilon^n}\right)_{\varepsilon=0} = L_W^n f\,(y,\, Y)\,.$$

Therefore, under the canonical mapping (6),

$$\bar{f}(y,\, Y;\, \varepsilon) = \sum_{n\geqslant 0}\frac{1}{n!}\,\varepsilon^n\left(\frac{\mathrm{d}^n\bar{f}}{\mathrm{d}\varepsilon^n}\right)_{\varepsilon=0} = \sum_{n\geqslant 0}\frac{1}{n!}\,\varepsilon^n L_W^n f\,(y,\, Y)\,,$$

and this conversion rule can be written concisely as

$$f\big(\exp(\varepsilon L_W)\,(y),\, \exp(\varepsilon L_W)\,(Y)\big) = \exp(\varepsilon L_W)\,\big(f\,(y,\, Y)\big)\,. \tag{7}$$

In particular

$$W(x(y, Y; \varepsilon), X(y, Y; \varepsilon)) = W(y, Y),$$

which can be interpreted by saying that the generator is 'conserved' under the canonical mapping (6).

If the function $f(x, X; \varepsilon)$ depends analytically on ε so that

$$f(x, X; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_n(x, X), \tag{8}$$

then its transform under (6) is

$$f(x(y, Y; \varepsilon), X(y, Y; \varepsilon); \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_W^m f_{n-m}(y, Y). \tag{9}$$

For, on applying rule (7) to each coefficient f_n, one obtains that

$$f_n(x(y, Y; \varepsilon), X(y, Y; \varepsilon)) = \sum_{m \geqslant 0} \frac{1}{m!} \varepsilon^m L_W^m f_n(y, Y);$$

then substituting these expressions into (8) and grouping like powers of ε establishes conversion rule (9).

As an application of rule (9), consider the transformation $(z, Z, \varepsilon) \to (y, Y)$ generated by the function $V(z, Z)$:

$$y = \exp(\varepsilon L_V)(z), \qquad Y = \exp(\varepsilon L_V)(Z). \tag{10}$$

In view of (9), the mapping $(z, Z, \varepsilon) \to (x, X)$ that is the composition of (6) and (10) can be written explicitly as follows:

$$\begin{aligned} x &= \sum_{n \geqslant 0} \frac{1}{\varepsilon^n} \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_V^m L_{W(z, Z)}^{n-m} z, \\ X &= \sum_{n \geqslant 0} \frac{1}{\varepsilon^n} \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_V^m L_{W(z, Z)}^{n-m} Z. \end{aligned} \tag{11}$$

These expressions are basic in establishing that the inverse of transformation (6) is

$$y = \exp(\varepsilon L_{-W(x, X)})(x), \qquad Y = \exp(\varepsilon L_{-W(x, X)})(X). \tag{12}$$

Indeed, make $V(x, X) = -W(x, X)$ in (11). According to (4_1),

$$L_{-W} L_W = L_W L_{-W},$$

so that

$$\sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_{-W}^m L_W^{n-m} = (L_{-W} + L_W)^n.$$

But $L_W + L_{-W} = 0$; thus, by taking $V(x, X) = -W(x, X)$ in the composition rules

(11), one obtains that $x=z$, $X=Z$, and this proves that (12) is the inverse of (6).

Definition (6) of a canonical mapping analytic in a small parameter extends to nonconservative transformations. This can be done in principle by observing what becomes of the expressions (6) when a moment is ignorable in the generator $W(y, Y)$.

In order to present these modifications in a form immediately suitable to routine calculations, consider the usual $(m+1)$-dimensional space of events (Synge, 1960) with coordinates v_j expressed in alternative notation as

$$v_j = y_j \quad (1 \leqslant j \leqslant m), \qquad v_{n+1} = t,$$

and a moment-energy space with components V_j to be expressed as

$$V_j = Y_j \quad (1 \leqslant j \leqslant m), \qquad V_{n+1} = -H.$$

It will be convenient to maintain the notation

$$L_W f = \sum_{1 \leqslant j \leqslant m} \left(\frac{\partial f}{\partial y_j} \frac{\partial W}{\partial Y_j} - \frac{\partial f}{\partial Y_j} \frac{\partial W}{\partial y_j} \right)$$

for the usual Poisson bracket in the phase space (y, Y), and to introduce the notation

$$\Lambda_W f = \sum_{1 \leqslant j \leqslant m+1} \left(\frac{\partial f}{\partial v_j} \frac{\partial W}{\partial V_j} - \frac{\partial f}{\partial V_j} \frac{\partial W}{\partial v_j} \right)$$

for the Poisson bracket in the space (v, V) of states and energy.

Consider the canonical mapping in the space of states and energy

$$u = \exp(\varepsilon \Lambda_W)(v) \qquad U = \exp(\varepsilon \Lambda_W)(V) \tag{13}$$

generated from a function $W(v, V)$, and introduce the alternative notation

$$x_j = u_j \quad (1 \leqslant j \leqslant m), \qquad s = u_{n+1},$$
$$X_j = U_j \quad (1 \leqslant j \leqslant m), \qquad K = -U_{n+1}.$$

Assume that the generator W does not depend on H (i.e., $\partial W/\partial H = 0$ identically). Consequently, for any function $f(v, V)$,

$$\Lambda_W f = L_W f + \frac{\partial f}{\partial H} \frac{\partial W}{\partial t}. \tag{14}$$

Moreover, if f itself does not depend on H, then

$$\Lambda_W f = L_W f.$$

There results that, in the alternative notation, the transformation (13) takes on the form

$$x(y, Y, t; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n L^n_{W(y, Y; t)} y,$$

$$X(y, Y, t; \varepsilon) = \sum_{n \geq 0} \frac{1}{n!} \varepsilon^n L^n_{W\,(y, Y; t)} Y\,, \tag{15}$$

similar to that of (6). Also

$$\Delta_W t = 0\,,$$

hence definition (13) implies that

$$s = t\,.$$

At last, applying (14) to the function $f = H$, one readily obtains that

$$\Delta_W H = \partial W / \partial t$$

and more generally, by induction over n, that

$$\Delta^n_W H = L^{n-1}_W \frac{\partial W}{\partial t}\,.$$

Therefore definition (13) implies that

$$K = H + \sum_{n \geq 1} \frac{1}{n!} \varepsilon^n L^{n-1}_W \frac{\partial W}{\partial t}\,, \tag{16}$$

which means that

$$R(y, Y; t) = -\sum_{n \geq 1} \frac{1}{n!} \varepsilon^n L^{n-1}_{W\,(y, Y; t)} \frac{\partial}{\partial t} W(y, Y; t) \tag{17}$$

is the remainder function (Wintner, 1941) of the nonconservative canonical transformation (15) generated from $W(y, Y; t)$.

3. Lie Transforms as Canonical Mappings

The preceding developments are not valid unless W does not depend explicitly on the small parameter ε. However, this restriction can be discarded, provided suitable modifications are introduced in the formalism.

Given a function $W(y, Y; \varepsilon)$, define for any function $f(y, Y; \varepsilon)$ the application

$$f \to \Delta_W f = L_W f + \frac{\partial f}{\partial \varepsilon}\,.$$

It has properties similar to those of the operator L_W, namely that

$$\Delta_W(\alpha f + \beta g) = \alpha \Delta_W f + \beta \Delta_W g\,, \tag{18_1}$$

$$\Delta_W(fg) = f \cdot \Delta_W g + g \cdot \Delta_W f\,, \tag{19_1}$$

$$\Delta_W(f; g) = (\Delta_W f; g) + (f; \Delta_W g), \tag{20_1}$$

$$\Delta_V \Delta_W = \Delta_W \Delta_V + L_{(W; V)} + L_{(\partial W/\partial \varepsilon - \partial V/\partial \varepsilon)}\,.$$

In the same way, define successively

$$\Delta_W^2 f = \Delta_W(\Delta_W f), \quad \Delta_W^3 f = \Delta_W(\Delta_W^2 f),$$

and so on; for convenient notations, put

$$\Delta_W^0 f = f \quad \text{and} \quad \Delta_W^1 f = \Delta_W f .$$

Exactly like for the operator L_W, one easily checks that, for any $n \geqslant 0$,

$$\Delta_W^n(\alpha f + \beta g) = \alpha \Delta_W^n f + \beta \Delta_W^n g , \tag{18$_n$}$$

$$\Delta_W^n(fg) = \sum_{0 \leqslant m \leqslant n} \binom{n}{m} \Delta_W^m f \, \Delta_W^{n-m} g , \tag{19$_n$}$$

$$\Delta_W^n(f ; g) = \sum_{0 \leqslant m \leqslant n} \binom{n}{m} (\Delta_W^m f ; \Delta_W^{n-m} g). \tag{20$_n$}$$

In order to simplify the notations, the function of y and Y which results from making $\varepsilon = 0$ in the expression $\Delta^n_{W(y, Y; \varepsilon)} f(y, Y; \varepsilon)$ will be more simply written as $f^{(n)}(y, Y; 0)$.

Consider now the formal power series

$$E_W(f) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f^{(n)}(y, Y; 0). \tag{21}$$

The mapping

$$E_W : f \to E_W(f)$$

will be called the *Lie transform generated by W*.

In view of (18$_n$), (19$_n$) and (20$_n$), one sees immediately that

$$E_W(\alpha f + \beta g) = \alpha E_W(f) + \beta E_W(g), \tag{18$_E$}$$

$$E_W(f \cdot g) = E_W(f) \cdot E_W(g), \tag{19$_E$}$$

$$E_W(f ; g) = (E_W f ; E_W g). \tag{20$_E$}$$

This last formula serves in particular to establish that the formal power series

$$x = E_W(y) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n y^{(n)}(y, Y; 0),$$

$$X = E_W(Y) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n Y^{(n)}(y, Y; 0) \tag{22}$$

define a completely canonical mapping $(y, Y, \varepsilon) \to (x, X)$.

The transformation (22) admits a fundamental interpretation. If (x, X) is the solution starting from (y, Y) at the point $\varepsilon = 0$ for the canonical system

$$\frac{\mathrm{d}x}{\mathrm{d}\varepsilon} = \frac{\partial}{\partial X} W(x, X; \varepsilon), \quad \frac{\mathrm{d}X}{\mathrm{d}\varepsilon} = -\frac{\partial}{\partial x} W(x, X; \varepsilon), \tag{23}$$

then, because

$$\frac{dx}{d\varepsilon} = (x; W) = \Delta_W x, \quad \frac{d^2x}{d\varepsilon^2} = ((x; W); W) + \left(x; \frac{\partial W}{\partial \varepsilon}\right) = \Delta_W^2 x,$$

and so on, the expansion of the solution (x, X) in power series of ε result in the expressions (22).

More generally, for a function $f(x, X; \varepsilon)$, along the solution $(x(y, Y; \varepsilon), X(y, Y; \varepsilon))$ of Equation (23), one has that

$$df/d\varepsilon = \Delta_W f, \quad d^2 f/d\varepsilon^2 = \Delta_W^2 f,$$

and so on; consequently

$$f(x(y, Y; \varepsilon), X(y, X; \varepsilon); \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n (\Delta_W^n f)_{\varepsilon = 0},$$

which can be written symbolically as

$$f(E_W(y), E_W(Y); \varepsilon) = (E_W(f))(y, Y; \varepsilon). \tag{24}$$

This is the rule to convert a function f of x, X and ε into one with arguments y, Y and ε through the canonical mapping (22).

When both the generating function $W(y, Y; \varepsilon)$ and the function f are power series of ε, the coefficients $f^{(n)}(y, Y; 0)$ in (21) can be constructed in a simple recursive fashion.

Assuming that

$$W(y, Y; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n W_{n+1}(y, Y), \tag{25}$$

$$f(y, Y; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_n(y, Y), \tag{26}$$

and, for the sake of simple notations, putting

$$L_{W_p} = L_p \quad (p \geqslant 1),$$

one finds that

$$\frac{\partial}{\partial \varepsilon} f(y, Y; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_{n+1}(y, Y),$$

$$L_W f(y, Y; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_{m+1} f_{n-m}(y, Y).$$

Thus, if one represents $\Delta_W(f)$ by the formal power series

$$\Delta_W(f) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_n^{(1)}(y, Y),$$

one finds that, for $n \geqslant 0$,

$$f_n^{(1)}(y, Y) = f_{n+1}(y, Y) + \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_{m+1} f_{n-m}(y, Y). \tag{27}$$

Consequently, as one expects,

$$f^{(1)}(y, Y; 0) = f_0^{(1)}(y, Y) = f_1 + L_1 f_0 = f_1 + (f_0; W_1).$$

Repeating the construction for

$$\Delta_W^2(f) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_n^{(2)}(y, Y),$$

one finds that

$$f_n^{(2)}(y, Y) = f_{n+1}^{(1)}(y, Y) + \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_{m+1} f_{n-m}^{(1)}(y, Y), \tag{28}$$

so that

$$f^{(2)}(y, Y; 0) = f_0^{(2)}(y, Y) = f_1^{(1)} + L_1 f_0^{(1)} = f_1^{(1)} + L_1 f^{(1)}.$$

But, in view of (27),

$$f_1^{(1)} = f_2 + (f_1; W_1) + (f_0; W_2),$$

and the expression for $f_0^{(2)}$ in explicit form is

$$f_0^{(2)} = f_2 + 2(f_1; W_1) + (f_0; W_2) + ((f_0; W_1); W_1).$$

The recurrence extends to any n: thus, for

$$\Delta_W^k f = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n f_n^{(k)}(y, Y),$$

one finds that, for any $n \geqslant 0$,

$$f_n^{(k)}(y, Y) = f_{n+1}^{(k-1)} + \sum_{0 \leqslant m \leqslant n} \binom{n}{m} L_{m+1} f_{n-m}^{(k-1)}. \tag{29}$$

In particular

$$f^{(k)}(y, Y; 0) = f_0^{(k)}(y, Y; \varepsilon) = f_1^{(k-1)} + L_1 f^{(k-1)}.$$

The recursion (29) is easily visualized in Figure 1. For instance, assume that all the elements of the first three lines have been computed. The filling of the 4th row begins with the computation of $f_2^{(1)}$ which involves only the elements above and including f_3 on the diagonal passing through f_3:

$$f_2^{(1)} = f_3 + L_1 f_2 + 2L_2 f_1 + L_3 f_0 ;$$

then one is ready to compute $f_1^{(2)}$, which involves $f_2^{(1)}$ and the elements above it on the diagonal through $f_2^{(1)}$:

$$f_1^{(2)} = f_2^{(1)} + L_1 f_1^{(1)} + L_2 f_0^{(1)} ;$$

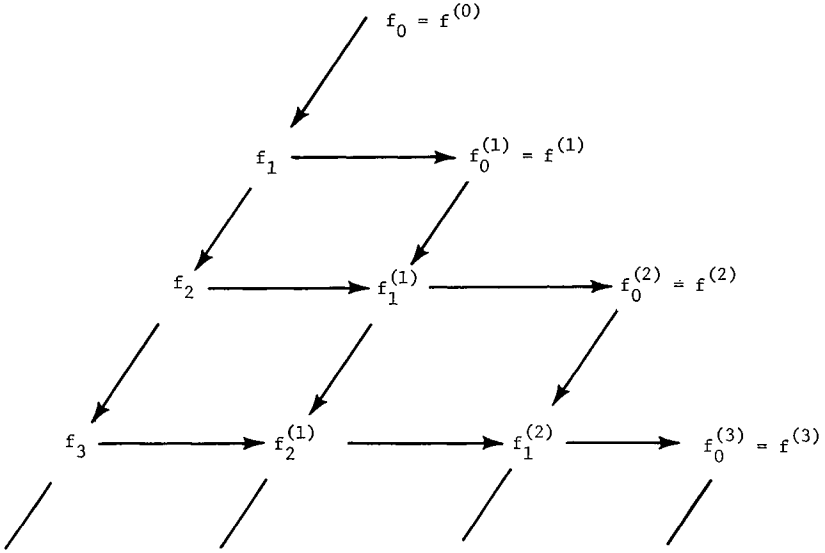


Fig. 1. Recursive transformation of an analytic function under a Lie transform.

the calculation ends with

$$f^{(3)} = f_0^{(3)} = f_1^{(2)} + L_1 f_0^{(2)}.$$

The construction is reminiscent of that of Pascal's triangle.

By applying the recurrence (29) to the series

$$f_{j,0} = y_j, \quad f_{j,n} = 0 \quad \text{for} \quad n \geqslant 1,$$
$$f_{j,0} = Y_j, \quad f_{j,n} = 0 \quad \text{for} \quad n \geqslant 1,$$

one easily constructs the coefficients of the various powers of ε in the formal series defining the canonical mapping (22). For the first few orders, one finds for instance that

$$\begin{aligned}
&y^{(1)} = L_1 y,\\
&y_1^{(1)} = L_2 y,\\
&y^{(2)} = y_1^{(1)} + L_1 y^{(1)},\\
&y_2^{(1)} = L_3 y,\\
&y_1^{(2)} = y_2^{(1)} + L_1 y_1^{(1)} + L_2 y^{(1)},\\
&y^{(3)} = y_1^{(2)} + L_1 y^{(2)},\\
&y_3^{(1)} = L_4 y,\\
&y_2^{(2)} = y_3^{(1)} + L_1 y_2^{(1)} + 2L_2 y_1^{(1)} + L_3 y^{(1)},\\
&y_1^{(3)} = y_2^{(2)} + L_1 y_1^{(2)} + L_2 y^{(2)},\\
&y^{(4)} = y_1^{(3)} + L_1 y^{(3)}.
\end{aligned}$$

The case of a nonconservative canonical mapping generated from a function

$W(y, Y, t; \varepsilon)$ can be treated as in Section 2 of this paper. On using the alternative notations introduced there, we define the differential operators

$$\Delta_W = L_W + \frac{\partial}{\partial \varepsilon}, \quad \nabla_W = \Delta_W + \frac{\partial}{\partial \varepsilon}.$$

Then, for any function $f(v, V)$,

$$\nabla_W f = \Delta_W f + \frac{\partial f}{\partial H} \cdot \frac{\partial W}{\partial t},$$

and therefore, if f does not depend explicitly on H,

$$\nabla_W f = \Delta_W f .$$

Hence, the transformation of the state variables can still be expressed by Equation (22):

$$x \equiv x(y, Y; t; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n y^{(n)}(y, Y; t; 0),$$

$$X \equiv X(y, Y; t; \varepsilon) = \sum_{n \geqslant 0} \frac{1}{n!} \varepsilon^n Y^{(n)}(y, Y; t; 0).$$

As in the case of a mapping defined by Lie series from a nonconservative generating function, we find that

$$\nabla_W t = 0, \quad \text{and} \quad \nabla_W H = \partial W / \partial t,$$

hence, by induction over n, that

$$\nabla_W^n H = \Delta_W^{n-1} \frac{\partial W}{\partial t}.$$

Consequently, the energy variable transforms according to the equation

$$K = H + \sum_{n \geqslant 1} \frac{1}{n!} \varepsilon^n \left(\Delta_W^{n-1} \frac{\partial W}{\partial t} \right)_{\varepsilon = 0},$$

which establishes that

$$R(y, Y; t; \varepsilon) = - \sum_{n \geqslant 1} \frac{1}{n!} \varepsilon^n \left(\Delta_W^{n-1} \frac{\partial W}{\partial t} \right)_{\varepsilon = 0} \tag{30}$$

is the remainder function of the nonconservative transformation. In the particular case when W is the power series (28), the preceding formula indicates that the remainder is to be constructed from the sequence $(\partial W_1/\partial t, \partial W_2/\partial t, \ldots, \partial W_n/\partial t, \ldots)$ by the recursion:

$$s_{n-1}^{(0)} = \partial W_n / \partial t,$$

$$s_j^{(i)} = s_{j+1}^{(i-1)} + \sum_{0 \leqslant k \leqslant j} \binom{j}{k} (s_{j-k}^{(i-1)}; W_{k+1}),$$

so that

$$R = -\sum_{n\geqslant 1} \frac{1}{n!}\varepsilon^n s_0^{(n-1)}(y, Y; t). \tag{31}$$

The inversion of the mapping (22) is a straightforward affair. Indeed (22) being essentially the flow defined by the Hamiltonian Equations (23), we know (Wintner, 1941) that its remainder function must be the function

$$R(y, Y; \varepsilon) = -W(x(y, Y; \varepsilon), X(y, Y; \varepsilon); \varepsilon), \tag{32}$$

and that under the mapping (22) the Equations (23) become

$$\mathrm{d}y/\mathrm{d}\varepsilon = \partial R/\partial Y, \quad \mathrm{d}Y/\mathrm{d}\varepsilon = -\partial R/\partial y. \tag{33}$$

Now that the inverse of (23) appears as the flow defined by (33), its generators turn out to be the coefficients of the various powers of ε in (32). But we obtain that expansion by applying the algorithm we just described for transforming a function $f(x, X; \varepsilon)$. To conclude, if W is the power series (25), we enter in the starting diagonal of Figure 1 the coefficients $-W_1(x, X; \varepsilon) - W_2(x, X; \varepsilon)$, and so on, and thus collect recursively the coefficients in the series

$$R(x, X; \varepsilon) = \sum_{n\geqslant 1} \frac{1}{n!}\varepsilon^n V_n(x, X). \tag{34}$$

For the first few steps, we easily see that

$$\begin{aligned} V_1(x, X) &= -W_1(x, X), \\ V_2(x, X) &= -W_2(x, X), \\ V_3(x, X) &= -W_3(x, X) - (W_2(x; X); W_1(x; X)), \\ V_4(x, X) &= -W_4 - 2(W_3; W_1) - ((W_2; W_1); W_1). \end{aligned} \tag{35}$$

4. Perturbation Theory Based on Lie Transforms

Consider the system of vector differential equations

$$\dot{x} = \mathscr{H}_X, \quad \dot{X} = -\mathscr{H}_x \tag{36}$$

derived from the Hamiltonian

$$\mathscr{H}(x, X; t; \varepsilon) = \mathscr{H}_0(x, X; t) + \varepsilon\mathscr{H}_1(x, X; t) + \frac{1}{2!}\varepsilon^2\mathscr{H}_2(x, X; t) + \frac{1}{3!}\varepsilon^3\mathscr{H}_3(x, X; t) + \cdots. \tag{37}$$

The essence of the technique proposed here consists as is customary in constructing a canonical transformation $(x, X, t, \varepsilon) \to (y, Y)$ analytic in ε at $\varepsilon = 0$ so as to achieve

in the transformed Hamiltonian

$$\mathscr{K}(y, Y; t; \varepsilon) = \mathscr{K}_0(y, Y; t) + \varepsilon \mathscr{K}_1(y, Y; t) + \frac{1}{2!}\varepsilon^2 \mathscr{K}_2(y, Y; t) + \frac{1}{3!}\varepsilon^3 \mathscr{K}_3(y, Y; t) + \cdots \tag{38}$$

specific requirements (suppression of periodic terms of assigned type, elimination of some or all angle coordinations, normalization, and so on). Where we differ from the usual von Zeipel's method is that we plan to build the mapping explicitly in the form of power series

$$\begin{aligned} x &= y + \varepsilon y^{(1)}(y, Y; t) + \frac{1}{2!}\varepsilon^2 y^{(2)}(y, Y; t) + \frac{1}{3!}\varepsilon^3 y^{(3)}(y, Y; t) + \cdots, \\ X &= Y + \varepsilon Y^{(1)}(y, Y; t) + \frac{1}{2!}\varepsilon^2 Y^{(2)}(y, Y; t) + \frac{1}{3!}\varepsilon^3 Y^{(3)}(y, Y; t) + \cdots, \end{aligned} \tag{39}$$

together with its inverse

$$\begin{aligned} y &= x + \varepsilon x^{(1)}(x, X; t) + \frac{1}{2!}\varepsilon^2 x^{(2)}(x, X; t) + \frac{1}{3!}\varepsilon^3 x^{(3)}(x, X; t) + \cdots, \\ Y &= X + \varepsilon X^{(1)}(x, X; t) + \frac{1}{2!}\varepsilon^2 X^{(2)}(x, X; t) + \frac{1}{3!}\varepsilon^3 X^{(3)}(x, X; t) + \cdots, \end{aligned} \tag{40}$$

and its remainder function

$$R(y, Y; t) = \varepsilon R^{(1)}(y, Y; t) + \frac{1}{2!}\varepsilon^2 R^{(2)}(y, Y; t) + \frac{1}{3!}\varepsilon^3 R^{(3)}(y, Y; t) + \cdots. \tag{41}$$

We shall describe in full detail the sequence of operations one has to perform in order to carry the transformation up to order 3 in ε.

The scheme is basically a recursive one; it is based on the principle that, with respect to the transformation (39), the canonical system (36) is equivalent to the system

$$\dot{y} = \mathscr{K}_Y, \qquad \dot{Y} = -\mathscr{K}_y \tag{42}$$

derived from the Hamiltonian

$$\mathscr{K}(y, Y; t; \varepsilon) = \mathscr{H}(x(y, Y; t; \varepsilon), X(y, Y; t; \varepsilon); t, \varepsilon) + R(y, Y; t). \tag{43}$$

As we impose that the mapping (39) be of the form

$$x = E_W(y), \qquad X = E_W(Y),$$

we have that

$$\mathscr{H}(x(y, Y; t; \varepsilon), X(y, Y; t; \varepsilon); t; \varepsilon) = E_W(\mathscr{H}).$$

Accordingly the target is now the construction of W and $\mathscr{K}$ to satisfy prescribed requirements in (42).

The recursion is entered by putting

$$\mathscr{K}_0(y, Y; t) = \mathscr{H}_0(y, Y; t). \tag{44}$$

The operations at order 1 in ε begins, as is obvious from the conversion rule established in Section 3, by considering the linear partial differential relation

$$(\mathscr{H}_0; W_1) - \frac{\partial W_1}{\partial t} + \mathscr{H}_1 = \mathscr{K}_1 \tag{45}$$

to be taken as an identity over the phase space (y, Y). Notice that $(\mathscr{H}_0; W_1)+\mathscr{H}_1$ is the coefficient $\mathscr{H}_0^{(1)}$ of ε in the Lie transform of $\mathscr{H}$, and that $-\partial W_1/\partial t$ is the coefficient of ε in the remainder of the transformation. Assuming that a certain selection has been made for $\mathscr{K}_1(y, Y; t)$, we then have in (43) a linear partial differential equation

$$\left(L_{\mathscr{H}_0} + \frac{\partial}{\partial t}\right) W_1 = \mathscr{H}_1 - \mathscr{K}_1, \tag{46}$$

which one solves to find the coefficient

$$W_1 \equiv W_1(y, Y; t)$$

of ε in the generating function. Then one computes that

$$\mathscr{H}_0^{(1)} \equiv \mathscr{H}_0^{(1)}(y, Y; t) = (\mathscr{H}_0; W_1) + \mathscr{H}_1, \tag{47}$$

$$y^{(1)} \equiv y^{(1)}(y, Y; t) = (y; W_1) = \frac{\partial W_1}{\partial Y}, \tag{48}$$

$$Y^{(1)} \equiv Y^{(1)}(y, Y; t) = (Y; W_1) = -\frac{\partial W_1}{\partial y}, \tag{49}$$

$$R^{(1)} \equiv R^{(1)}(y, Y; t) = -\frac{\partial W_1}{\partial t}. \tag{50}$$

As we have shown in Section 3, the coefficients of ε in the inverse mappings are obtained by replacing the letters y and Y by x and X respectively in the coefficients $y^{(1)}$ and $Y^{(1)}$, and putting

$$\begin{aligned} x^{(1)} &\equiv x^{(1)}(x, X; t) = -y^{(1)}, \\ X^{(1)} &\equiv X^{(1)}(x, X; t) = -Y^{(1)}. \end{aligned} \tag{51}$$

At order 2 in ε, the provisional functions

$$\begin{aligned} \tilde{\mathscr{H}}_1^{(1)} &\equiv \tilde{\mathscr{H}}_1^{(1)}(y, Y; t) = \mathscr{H}_2 + (\mathscr{H}_1; W_1), \\ \tilde{\mathscr{H}}_0^{(2)} &\equiv \tilde{\mathscr{H}}_0^{(2)}(y, Y; t) = \tilde{\mathscr{H}}_1^{(1)} + (\tilde{\mathscr{H}}_0^{(1)}; W_1), \\ \tilde{R}^{(2)} &\equiv \tilde{R}^{(2)}(y, Y; t) = (R^{(1)}; W_1) \end{aligned} \tag{52}$$

are prepared. The tildes surmounting the coefficients are there to remind us that, for instance, $\tilde{\mathscr{H}}_0^{(2)}$ is only part of the coefficient of ε^2 in the Lie transform of $\mathscr{H}$: the complete coefficient will be obtained in (57) after W_2 has been computed. In view of

(43), we set up the partial differential relation

$$(\mathscr{H}_0; W_2) + \tilde{\mathscr{H}}_0^{(2)} - \frac{\partial W_2}{\partial t} + \tilde{R}^{(2)} = \mathscr{K}_2. \tag{53}$$

The first two terms in the left hand member are the coefficient of ε^2 in $E_W(\mathscr{H})$, the last two terms the coefficient of ε^2 in the remainder. The unknown function $\mathscr{K}_2$ is selected in compliance with the goals proposed for the transformation, and the resulting linear partial differential equation

$$\left(L_{\mathscr{H}_0} + \frac{\partial}{\partial t}\right) W_2 = \tilde{\mathscr{H}}_0^{(2)} + \tilde{R}^{(2)} - \mathscr{K}_2 \tag{54}$$

is integrated to yield the coefficient $W_2(y, Y; t)$ of ε in the generating function. The step of second order is completed by computing

$$\begin{aligned}
y_1^{(1)} &\equiv y_1^{(1)}(y, Y; t) = \frac{\partial W_2}{\partial Y}, \\
y^{(2)} &\equiv y^{(2)}(y, Y; t) = y_1^{(1)} + (y^{(1)}; W_1), \\
Y_1^{(1)} &\equiv Y_1^{(1)}(y, Y; t) = -\frac{\partial W_2}{\partial y}, \\
Y^{(2)} &\equiv Y^{(2)}(y, Y; t) = Y_1^{(1)} + (Y^{(1)}; W_1), \\
R^{(2)} &\equiv R^{(2)}(y, Y; t) = \tilde{R}^{(2)} - \frac{\partial W_2}{\partial t}.
\end{aligned} \tag{55}$$

The coefficients of ε^2 in the inverse mapping are found by replacing the letters y and Y by x and X respectively in the functions $y_1^{(1)}$, $Y_1^{(1)}$, $(y^{(1)}; W_1)$, $(Y^{(1)}; W_1)$, and then by putting

$$\begin{aligned}
x^{(2)} &\equiv x^{(2)}(x, X; t) = -y_1^{(1)} + (y^{(1)}; W_1), \\
X^{(2)} &\equiv X^{(2)}(x, X; t) = -Y_1^{(1)} + (Y^{(1)}; W_1).
\end{aligned} \tag{56}$$

We prepare the operations at order 3 by first completing the coefficients of order 2 in $E_W(\mathscr{H})$:

$$\begin{aligned}
\mathscr{H}_1^{(1)} &\equiv \mathscr{H}_1^{(1)}(y, Y; t) = \tilde{\mathscr{H}}_1^{(1)} + (\mathscr{H}_0; W_2), \\
\mathscr{H}_0^{(2)} &\equiv \mathscr{H}_0^{(2)}(y, Y; t) = \tilde{\mathscr{H}}_0^{(2)} + (\mathscr{H}_0; W_2),
\end{aligned} \tag{57}$$

and then, by computing the provisional elements

$$\begin{aligned}
\tilde{\mathscr{H}}_2^{(1)} &\equiv \tilde{\mathscr{H}}_2^{(1)}(y, Y; t) = \mathscr{H}_3 + (\mathscr{H}_2; W_1) + 2(\mathscr{H}_1; W_2), \\
\tilde{\mathscr{H}}_1^{(2)} &\equiv \tilde{\mathscr{H}}_1^{(2)}(y, Y; t) = \tilde{\mathscr{H}}_2^{(1)} + (\mathscr{H}_1^{(1)}; W_1) + (\mathscr{H}_0^{(1)}; W_2), \\
\tilde{\mathscr{H}}_0^{(3)} &\equiv \tilde{\mathscr{H}}_0^{(3)}(y, Y; t) = \tilde{\mathscr{H}}_1^{(2)} + (\mathscr{H}_0^{(2)}; W_1), \\
\tilde{R}_1^{(2)} &\equiv \tilde{R}_1^{(2)}(y, Y; t) = (R^{(1)}; W_2) - \left(\frac{\partial W_2}{\partial t}; W_1\right), \\
\tilde{R}^{(3)} &\equiv \tilde{R}^{(3)}(y, Y; t) = \tilde{R}_1^{(2)} + (R^{(2)}; W_1).
\end{aligned}$$

From the linear partial differential relation

$$(\mathscr{H}_0; W_3) + \tilde{\mathscr{H}}_0^{(3)} + \tilde{R}^{(3)} - \frac{\partial W_3}{\partial t} = \mathscr{K}_3,$$

the unknown function is selected, and the resulting linear partial differential equation

$$\left(L_{\mathscr{H}_0} + \frac{\partial}{\partial t}\right) W_3 = \tilde{\mathscr{H}}_0^{(3)} + \tilde{R}^{(3)} - \mathscr{K}_3$$

is solved to yield the coefficient $W_3(y, Y; t)$ of order 3 in the generating function. Then the sequence of operations

$$\begin{aligned}
y_2^{(1)} &= \frac{\partial W_3}{\partial Y}, \\
y_1^{(2)} &= y_2^{(1)} + (y_1^{(1)}; W_1) + (y_0^{(1)}; W_2), \\
y^{(3)} &= y_1^{(2)} + (y_0^{(2)}; W_1), \\
Y_2^{(1)} &= -\frac{\partial W_3}{\partial y}, \\
Y_1^{(2)} &= Y_2^{(1)} + (Y_1^{(1)}; W_1) + (Y_0^{(1)}; W_2), \\
Y_0^{(3)} &= Y_1^{(2)} + (Y_0^{(2)}; W_1)
\end{aligned}$$

will produce the 3rd order coefficients in the canonical mapping, whereas

$$R^{(3)} = \tilde{R}^{(3)} - \partial W_3/\partial t$$

is the 3rd order coefficient in the remainder function.

We explained at the end of Section 3 how to compute the coefficient of third order in the function generating the inverse mapping.

Clearly, if need be, the entire procedure can be extended to any order $N>3$.

5. Conclusions

Canonical mappings defined by Lie transforms as formal power series of a small parameter constitute the natural ingredient of Transformation Theory applied to Hamiltonian systems.

The formalism we just described is now being developed in four areas: a revision of Brouwer's theory of an artificial satellite, its extension to Lunar Orbiters, Birkhoff's normalization of the triangular centers of libration in the reduced problem of three bodies, and the analysis of resonances.

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